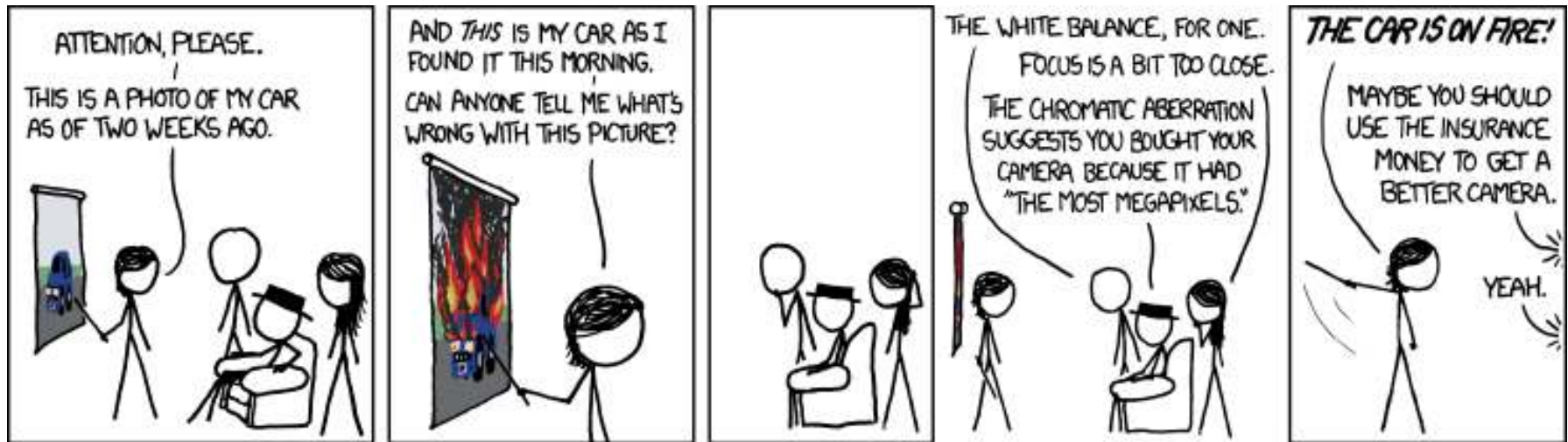


Computational Cameras



Lecture 0: Computational Cameras

Introduction & Syllabus

Instructor: Katie Bouman

TAs: Brandon Zhao, Brady Bhalla, Zachary Huang

Overview of today's lecture

- Teaching staff introductions
- Course logistics
- Evaluation & Collaboration
- What are computational cameras?
- Course Schedule & Invited Speakers
- Homework & Supplies

Teaching staff introductions

Instructor : Prof. Katherine (Katie) Bouman

Email: klbouman@caltech.edu

Office hours: Wednesday 1:30-3 PT
(see syllabus for exceptions)

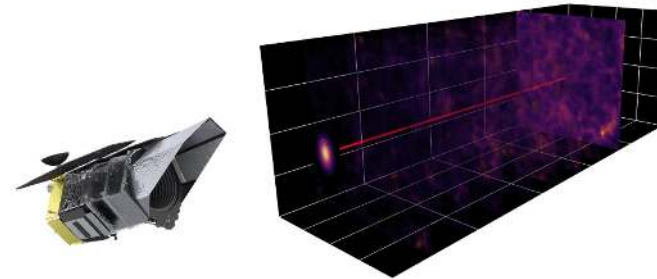
My group combines ideas from signal processing, computer vision, machine learning, and physics to design computational imaging systems. These systems tightly integrate algorithm and sensor design, making it possible to observe phenomena previously difficult or impossible to measure with traditional approaches



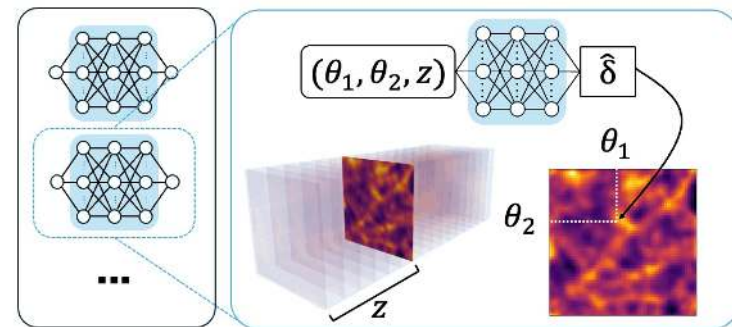
Teaching Assistant: Brandon Zhao

Email: byzhao@caltech.edu

Office hours: Thursday 3-4:30



Observed dark matter lensing

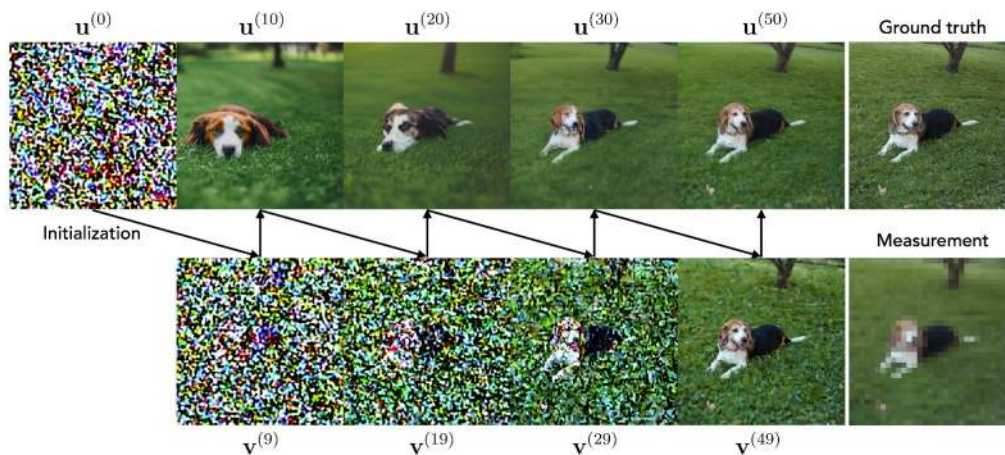


Teaching Assistant: Brady Bhalla

Email: bbhalla@caltech.edu

Office hours: Friday 4-5 pm

- Senior studying CS
- Took this course in 2024

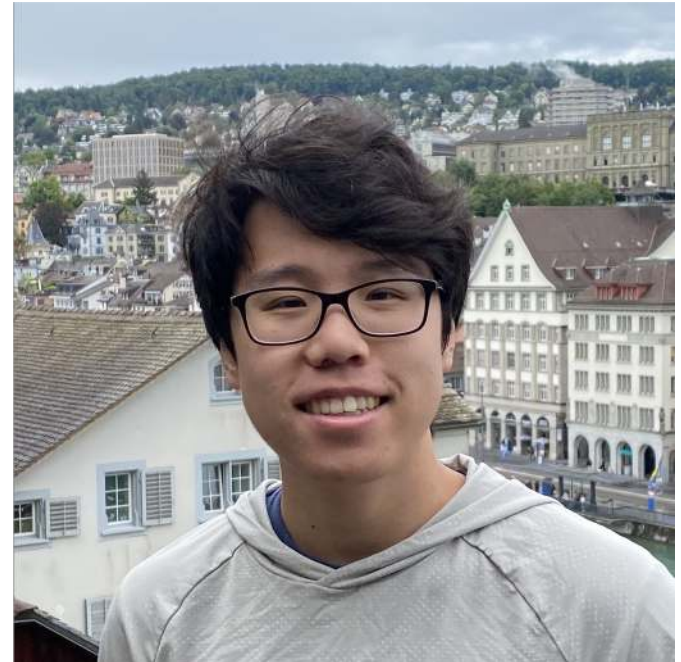
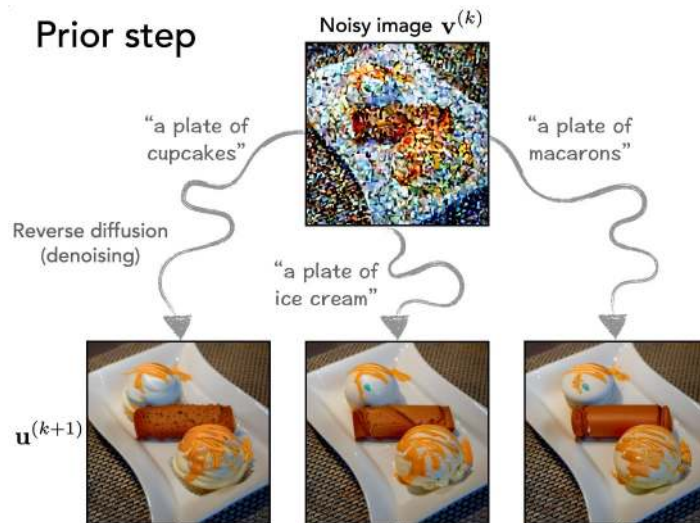


Teaching Assistant: Zachary Huang

Email: zwhuang@caltech.edu

Office hours: Monday 4-5:30

- Senior studying CS
- Took this course in 2024



Course logistics

- Course website on Canvas:

<https://caltech.instructure.com/courses/9727/pages/cs-166-front-page>

- Piazza Discussion Page (meant primarily for student-to-student interactions):
- GradeScope for homework submissions (Entry Code V4X5YN)

Course logistics

- Lectures will be held Tuesdays and Thursdays 1:00 PM – 2:25 PM PT
- Lecture notes will be posted on canvas
- Participate!



Office Hours

Katie: Wednesday 1:30-3:00 PT in Annenberg 346

Brandon: Thursday 3-4:30 in Annenberg 104

Brady: Friday 4-5 in Annenberg 104

Zachary: Monday 4-5:30 in Annenberg 104

homework is due and released on Tuesday
(except the first assignment!)

Questions?

Evaluation

- Five homework assignments (65%):
 - programming and capturing your own photographs with a Raspberry Pi
 - most programming will be in python.
 - five late days, you can use them as you want. After which, late homework will be penalized by 20% per day
 - conceptual discussion is allowed, but your solution code should be written and read by you and you alone.
- Final project (30%):
 - we will provide more information near the end of the course.
 - no exam, but final project presentations are during the exam period.
 - cannot use late days
- Participation (5%):
 - participate in lectures, and/or Piazza discussions, and/or office hours

Collaboration Policy

Basic discussion of the problems	Yes
Look at communal materials while writing up solutions	No
Look at other's non-communal work (i.e., writeups, code)	No
Turn in a set with more than one name on it (<u>does</u> not include naming collaborators in basic discussion)	No
Use of LLM and/or AI agent for help in assignments	No on homework May be used with instructor permission in the final project

Comments: Conceptual discussion is allowed, but your solution code should be written and read by you and you alone. Additionally, you may not look at course material from other schools or post your answers online (this includes e.g. public GitHub repositories).

Do not use LLM/AI Agent for help in assignments!

Academic Integrity: Abide by the Caltech Honor Code. All instances of plagiarism or other academic misconduct will be referred to the Board of Control for review. To find out more about understanding and avoiding plagiarism: <http://writing.caltech.edu/resources/plagiarism>.

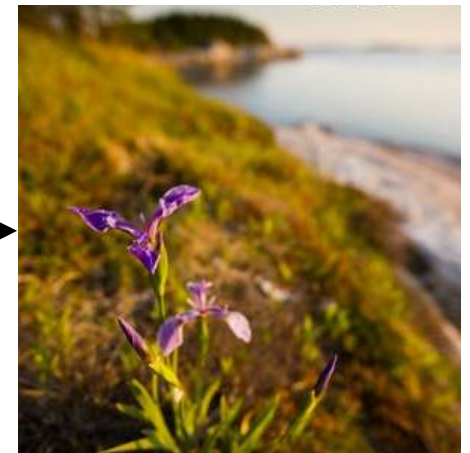
Contact me ASAP with any needs

- Students who may need an academic accommodation based on the impact of a disability must initiate the request with Caltech Accessibility Services for Students (CASS).
- If there are personal or health reasons you cannot attend lecture let me know
- If you are colorblind let me know, as it may affect a problem on the first homework assignment.

Questions?

What are Computational Cameras?

Traditional Photography

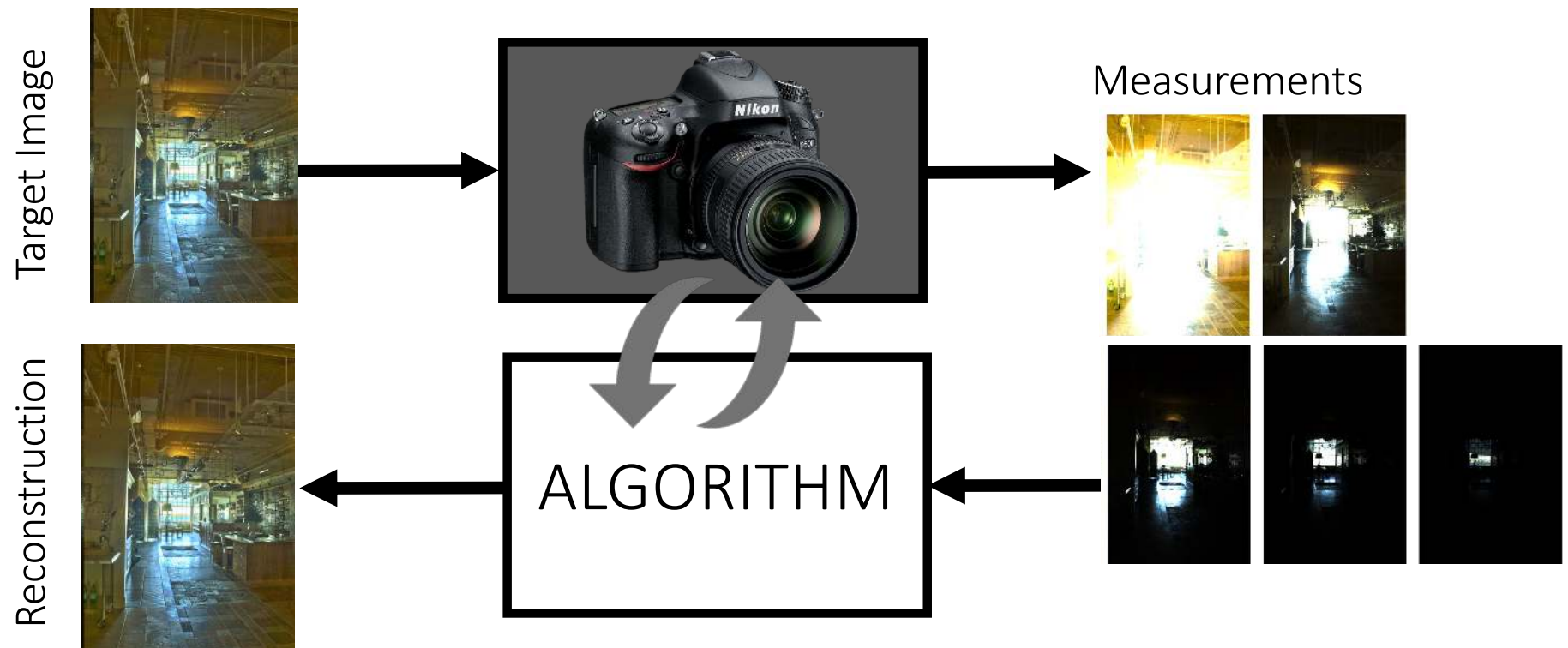


Traditional Photography

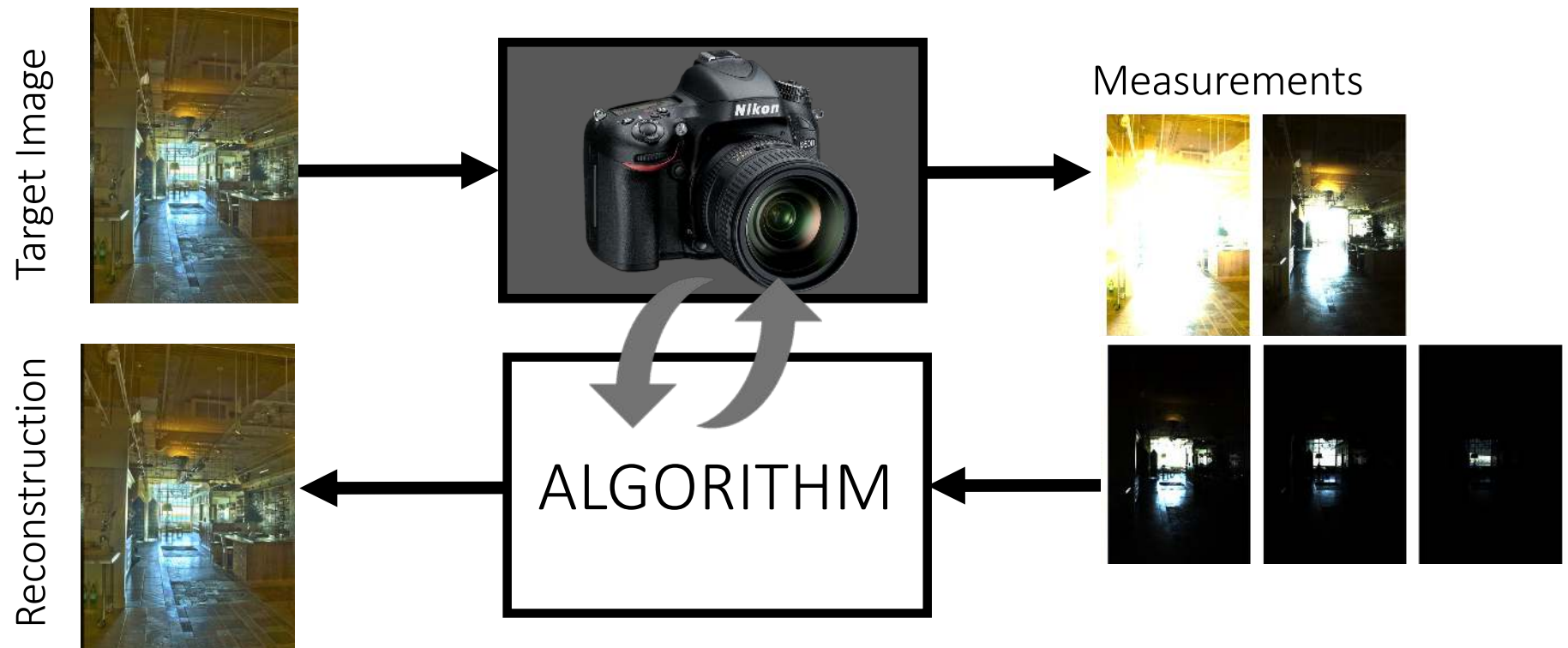
Target Image

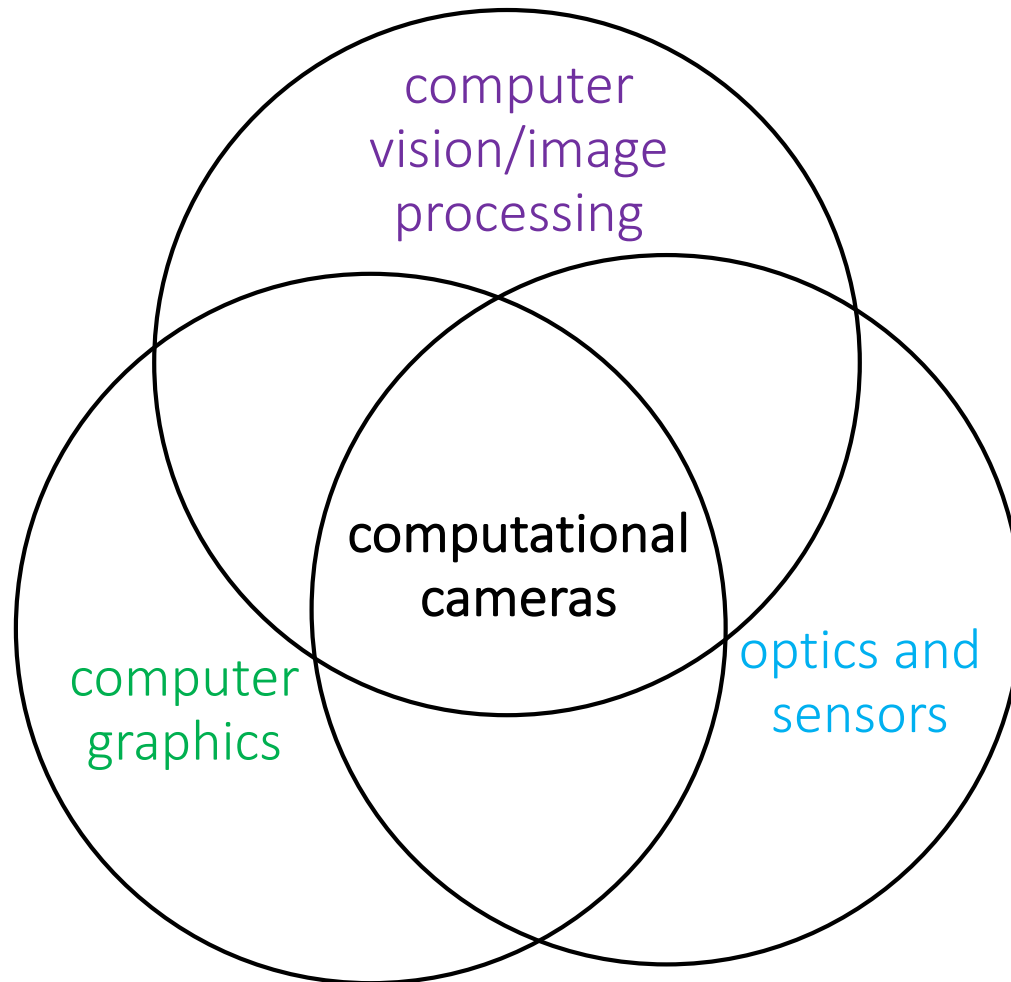


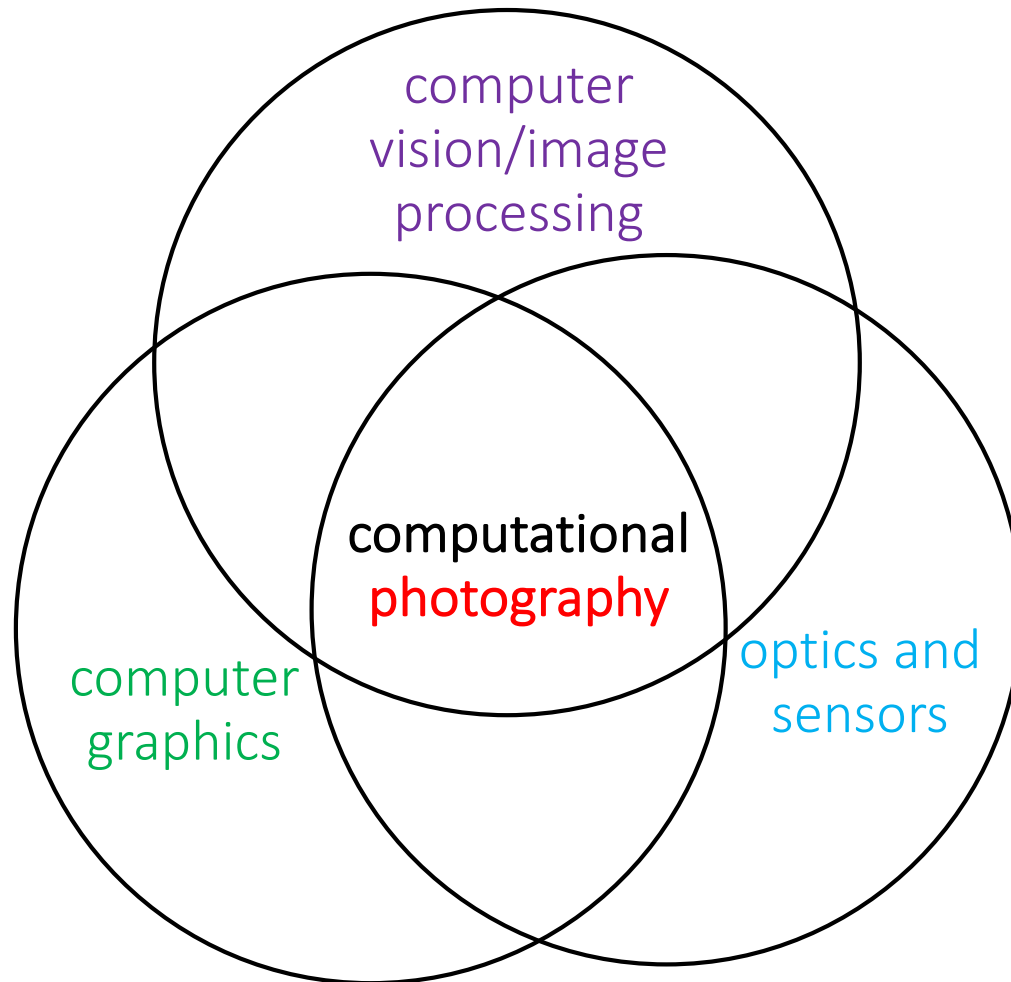
Unconventional Photography

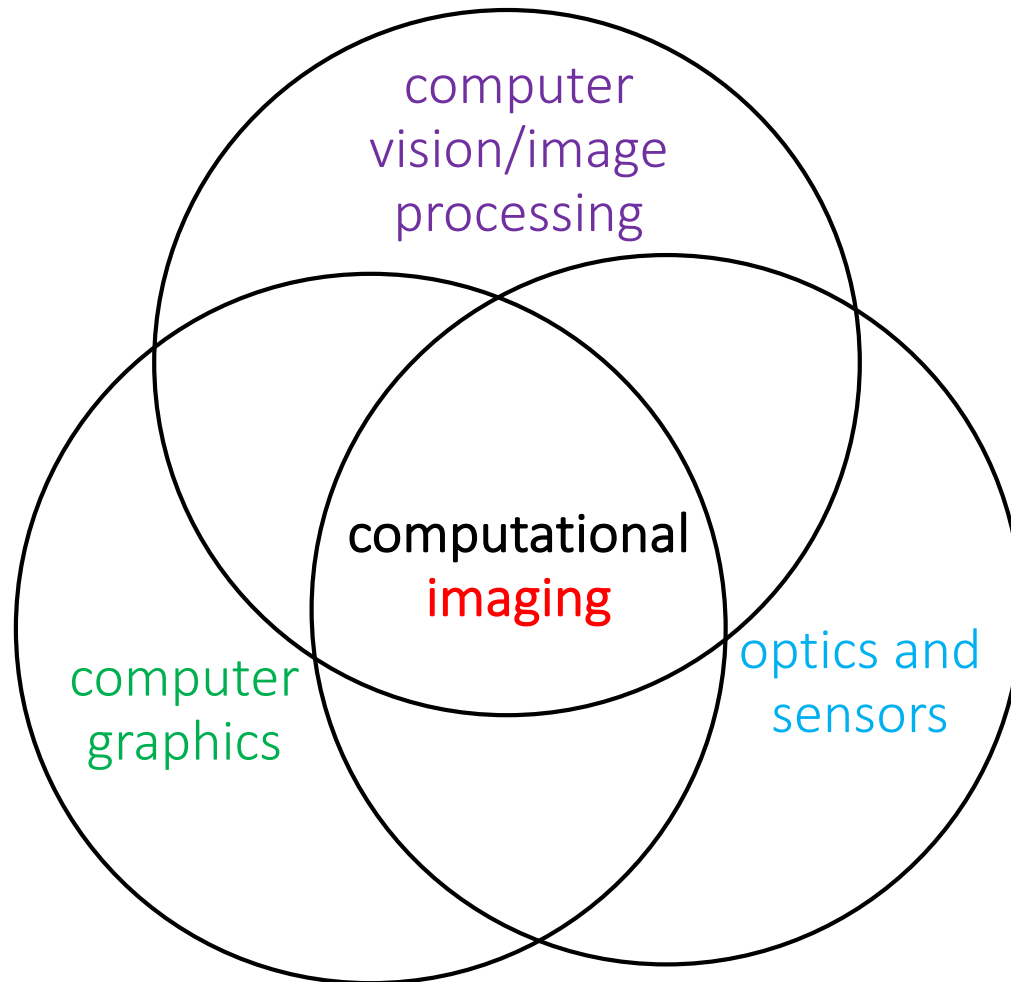


Computational Cameras









Analog Photography



optics to focus light on
an image plane



film to capture focused light
(chemical process)



dark room for limited post-
processing (chemical process)

Digital Photography



optics to focus light on
an image plane



digital sensor to capture focused
light (electrical process)



on-board processor for post-
processing (digital process)

Computational Cameras



optics to focus light on
an image plane



digital sensor to capture focused
light (electrical process)



arbitrary computation
between sensor and image

Imaging Enhancement for a Photographic Look



camera output

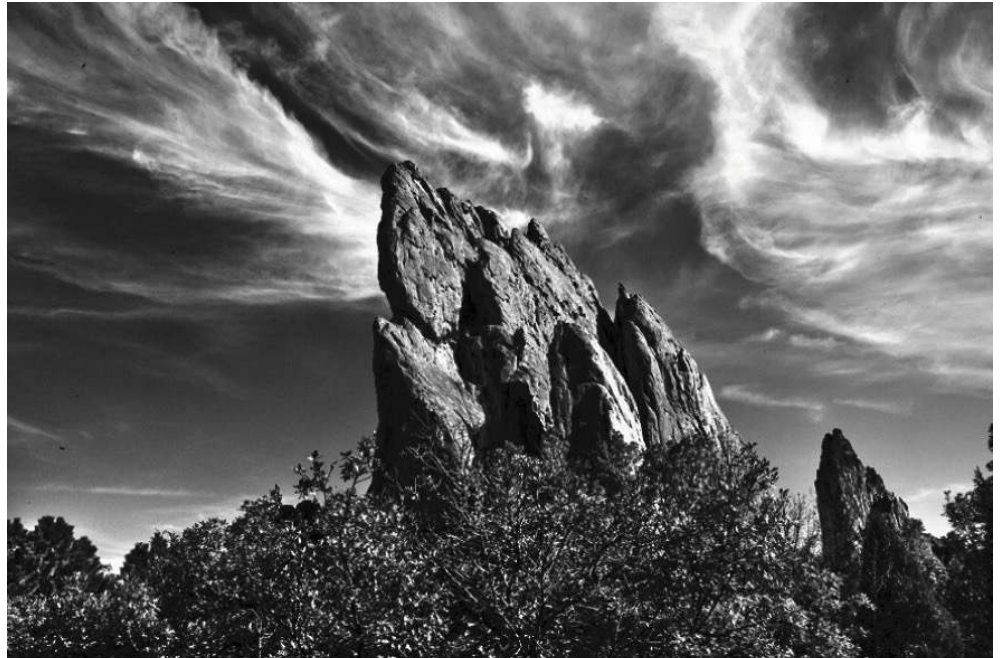


image after stylistic tonemapping

[Bae et al., SIGGRAPH 2006]

Style Transfer for Headshot Portraits



(a) Input: a casual face photo

(b) Outputs: new headshots with the styles transferred from the examples. The insets show the examples.

Learning Based Tonemapping



12 megapixel 16-bit linear input
(tone-mapped for visualization)



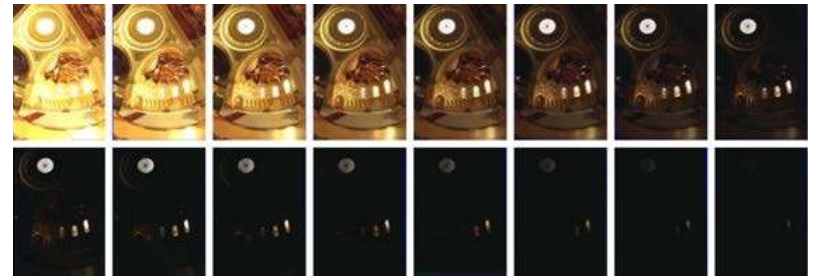
processed with our algorithm
61 ms, PSNR = 28.4 dB

[Gharbi, et al., 2017]

High Dynamic Range (HDR) Photography

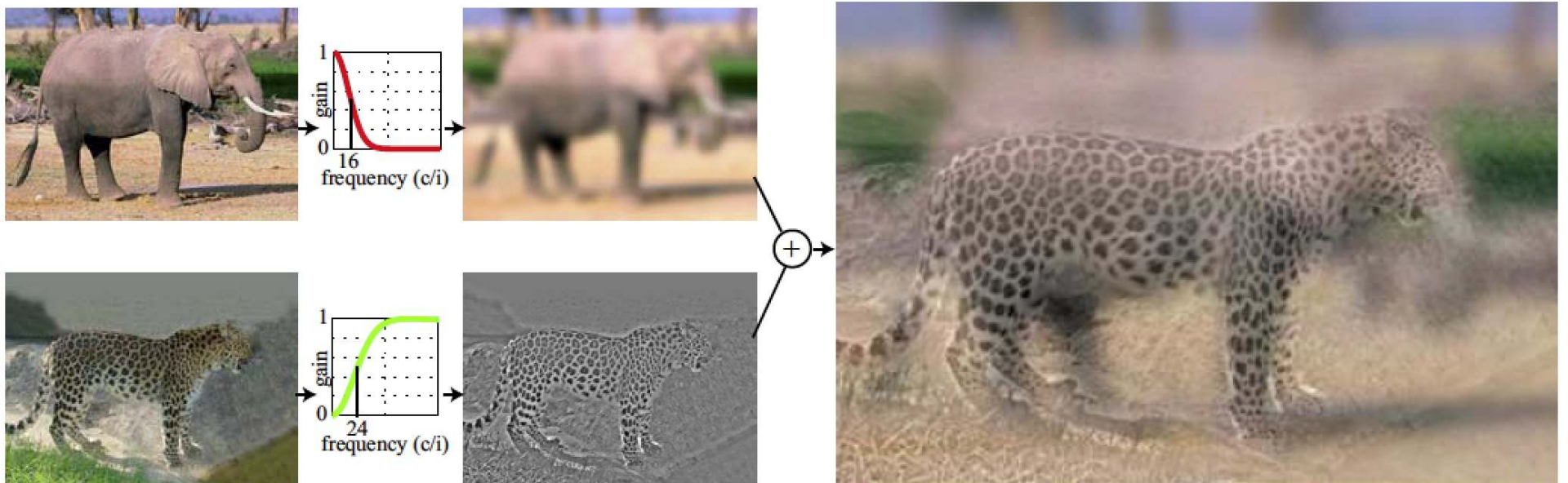


One of your
homeworks!

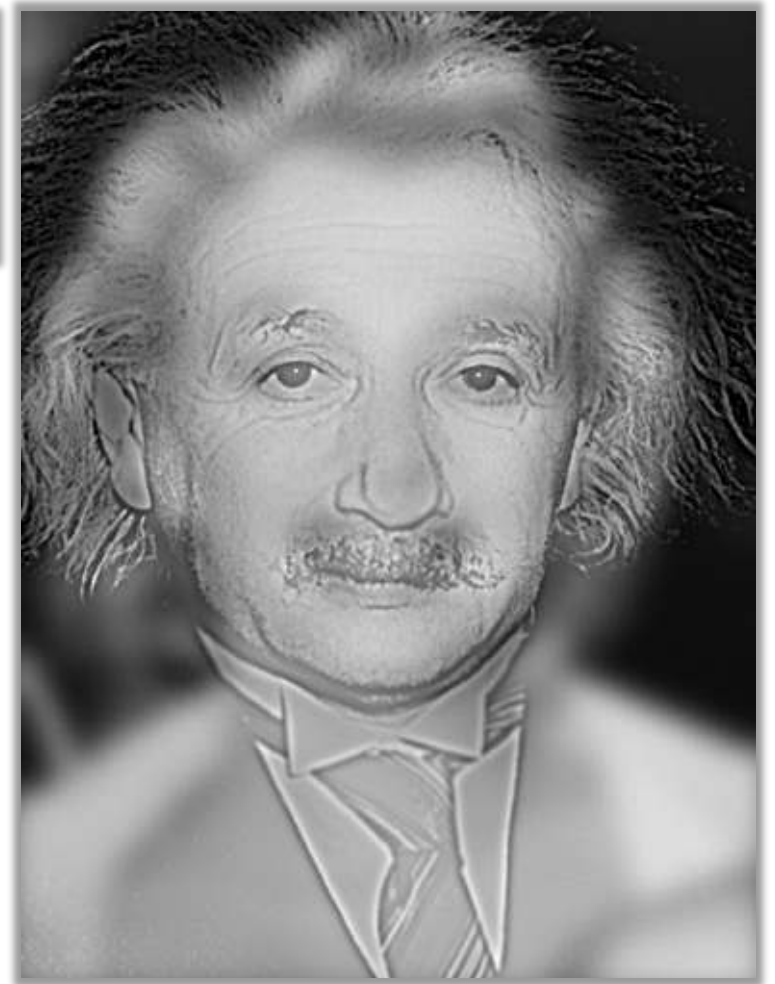


[Debevec and Malik, SIGGRAPH 1997]

Hybrid Images

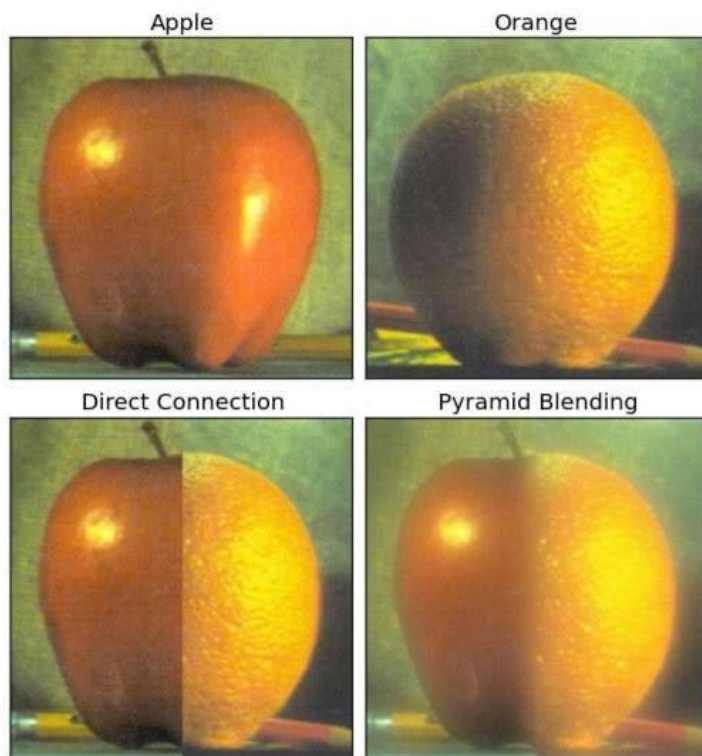


[Oliva SIGGRAPH 2006]



[Oliva SIGGRAPH 2006]

Image blending



One of your
homeworks!

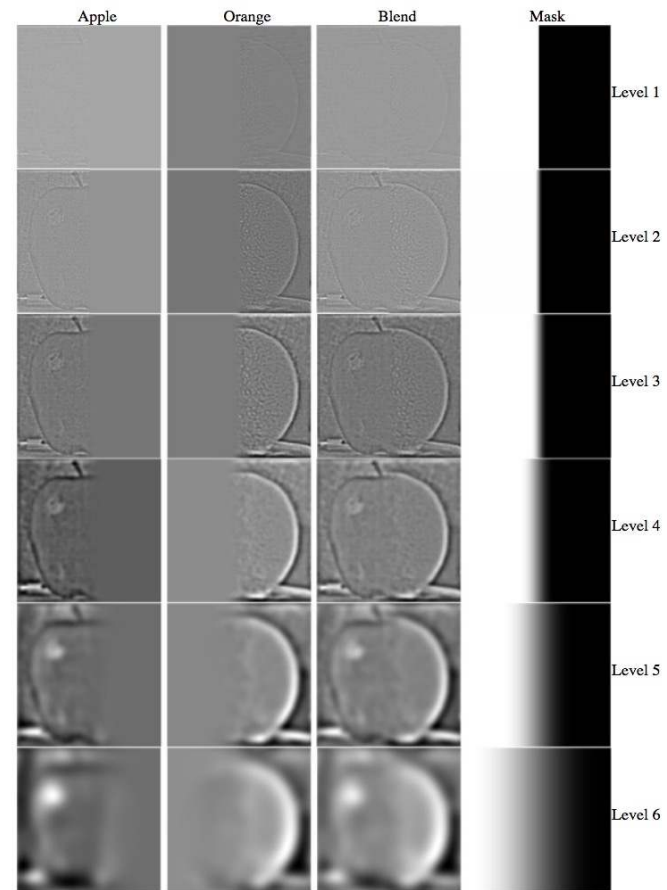
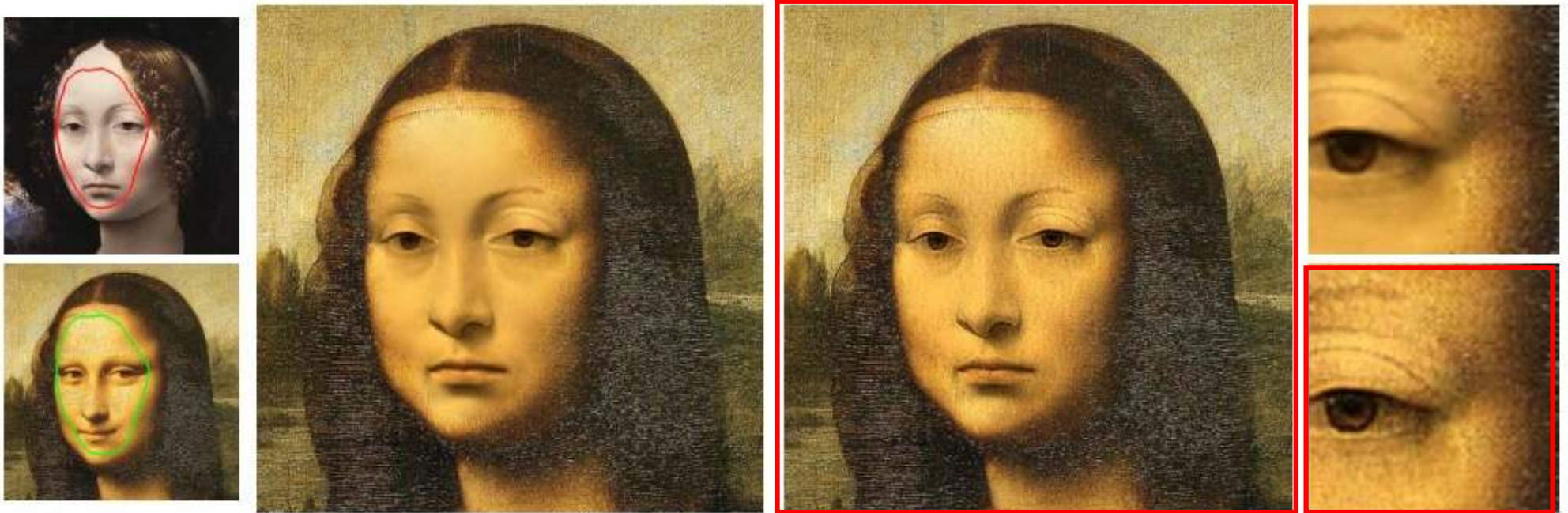
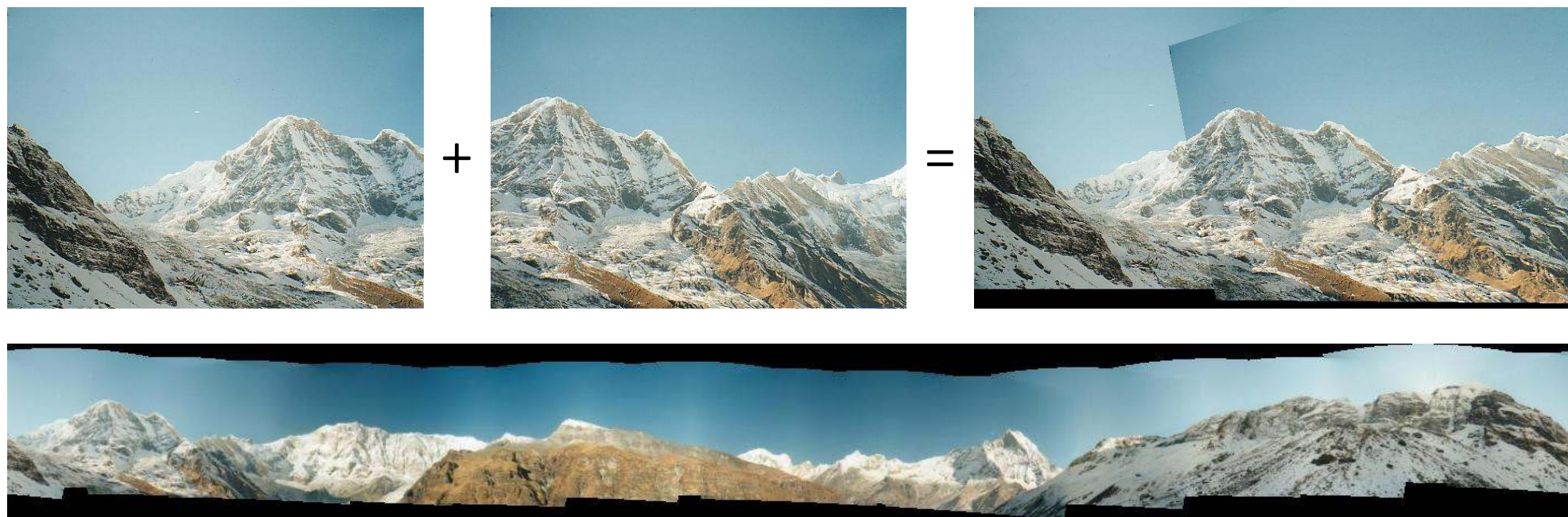


Image blending



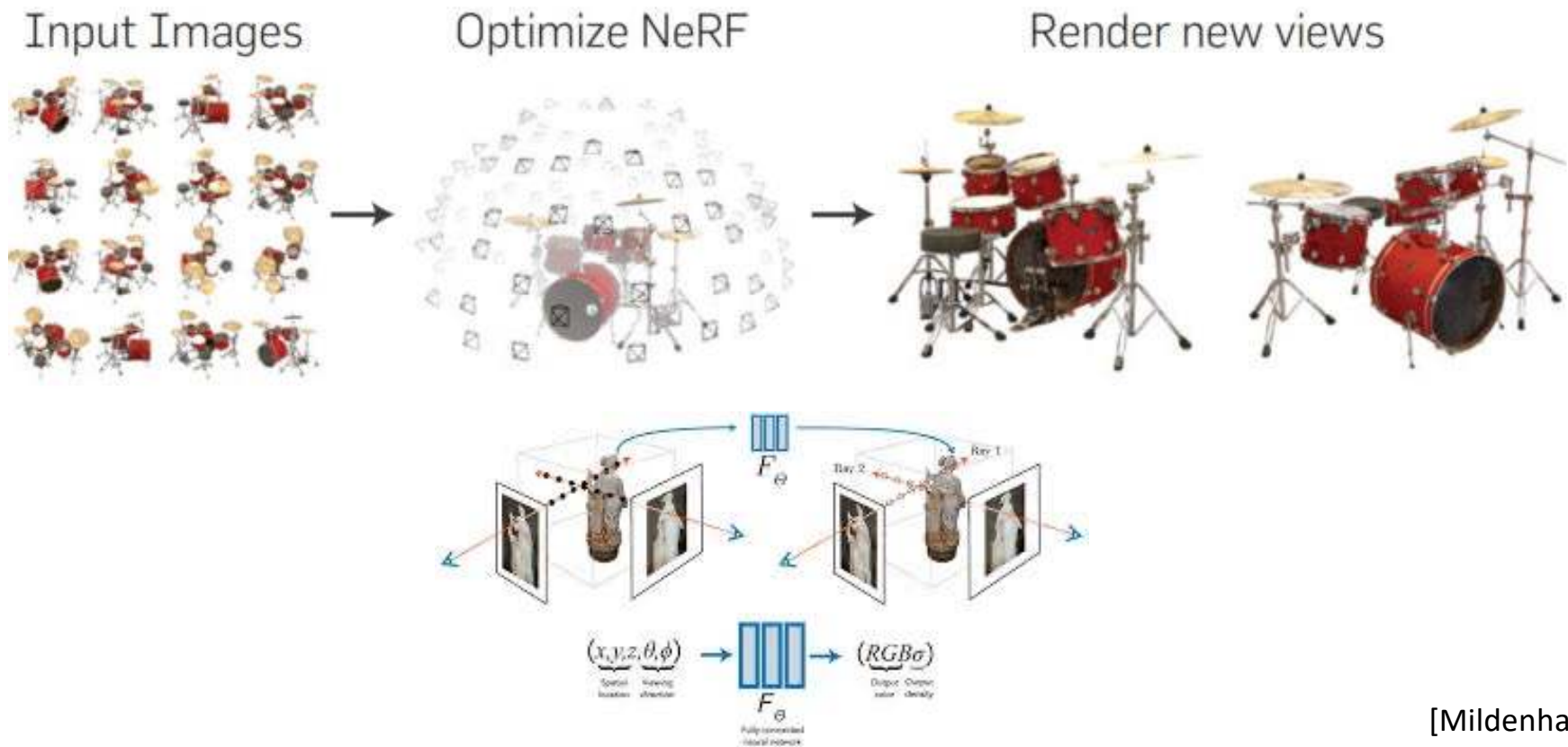
[Sunkavalli et al., SIGGRAPH 2010]

Auto-Stitching Images into Panoramas

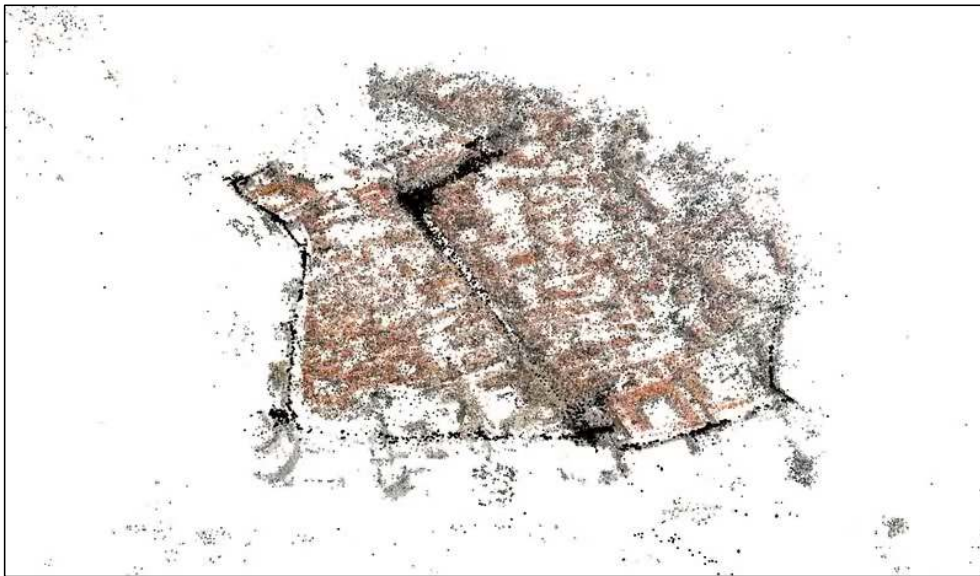


[Brown and Lowe, IJCV 2007]

3D Structure from Collections of Images



Organizing images to find structure



reconstructing cities from Internet photos



time-lapse from Internet photos

[Agarwal et al., ICCV 2009] [Martin-Brualla et al., SIGGRAPH 2015]

Modifying images to find structure



[Rubinstein, CVPR 2011]

Modifying images to find structure



[Wu, SIGGRAPH 2012]

Modifying images to find structure



[Wu, SIGGRAPH 2012]

Computational Cameras



optics to focus light on
an image plane



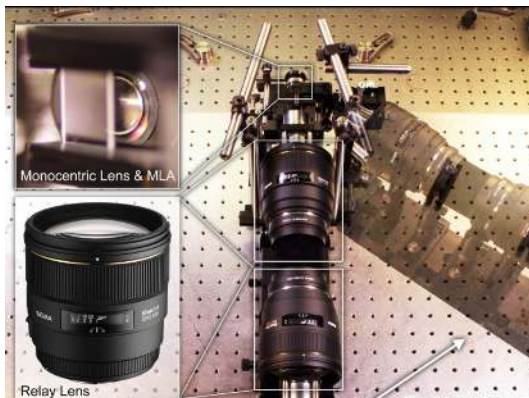
digital sensor to capture focused
light (electrical process)



arbitrary computation
between sensor and image

Slide courtesy of Ioannis Gkioulekas

Computational Cameras



generalized optics
between scene and sensor



digital sensor to capture focused
light (electrical process)



arbitrary computation
between sensor and image

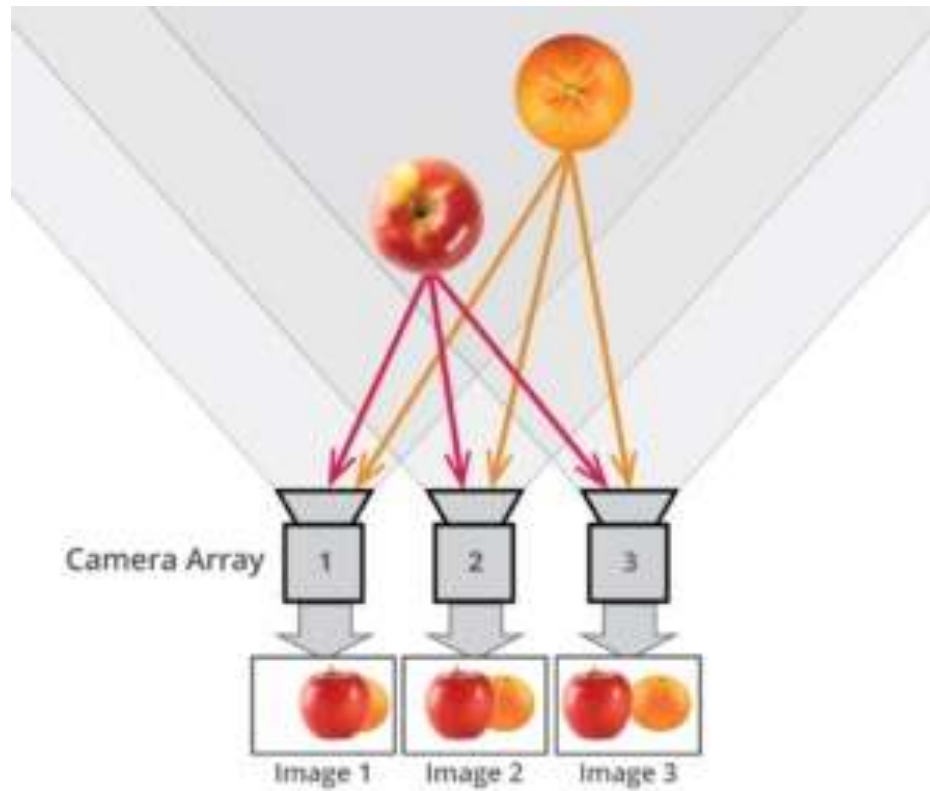
Slide courtesy of Ioannis Gkioulekas

Light Fields



Lytro

Light Fields

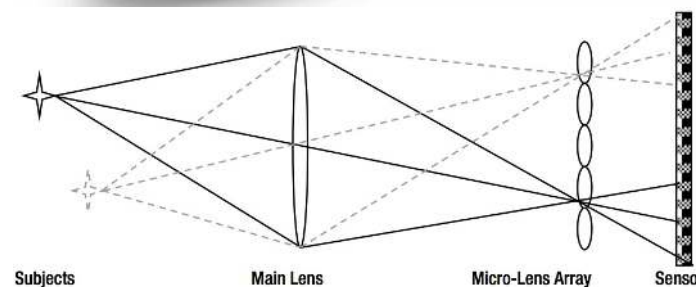


Light Field Cameras for Plenoptic Imaging



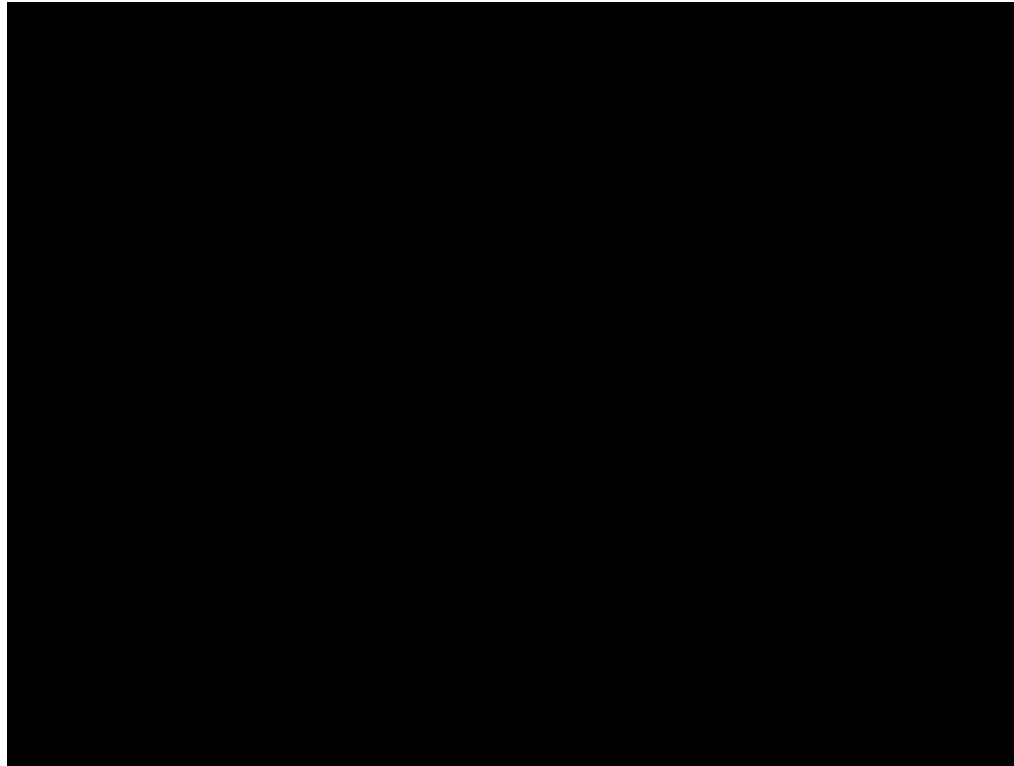
post-capture refocusing

One of your
homeworks!



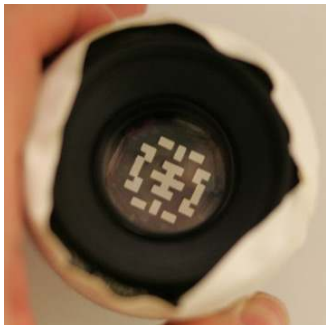
[Ng et al., SIGGRAPH 2005] [Lytro Inc.]

Light Field Cameras for Plenoptic Imaging

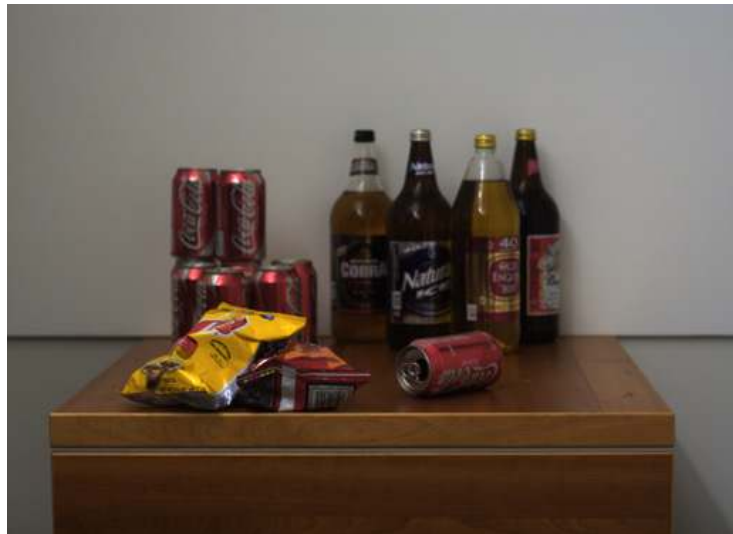


[Ng et al., SIGGRAPH 2005] [Lytro Inc.]

Coded Aperture for Single-Image Depth



conventional vs
coded lens



input image

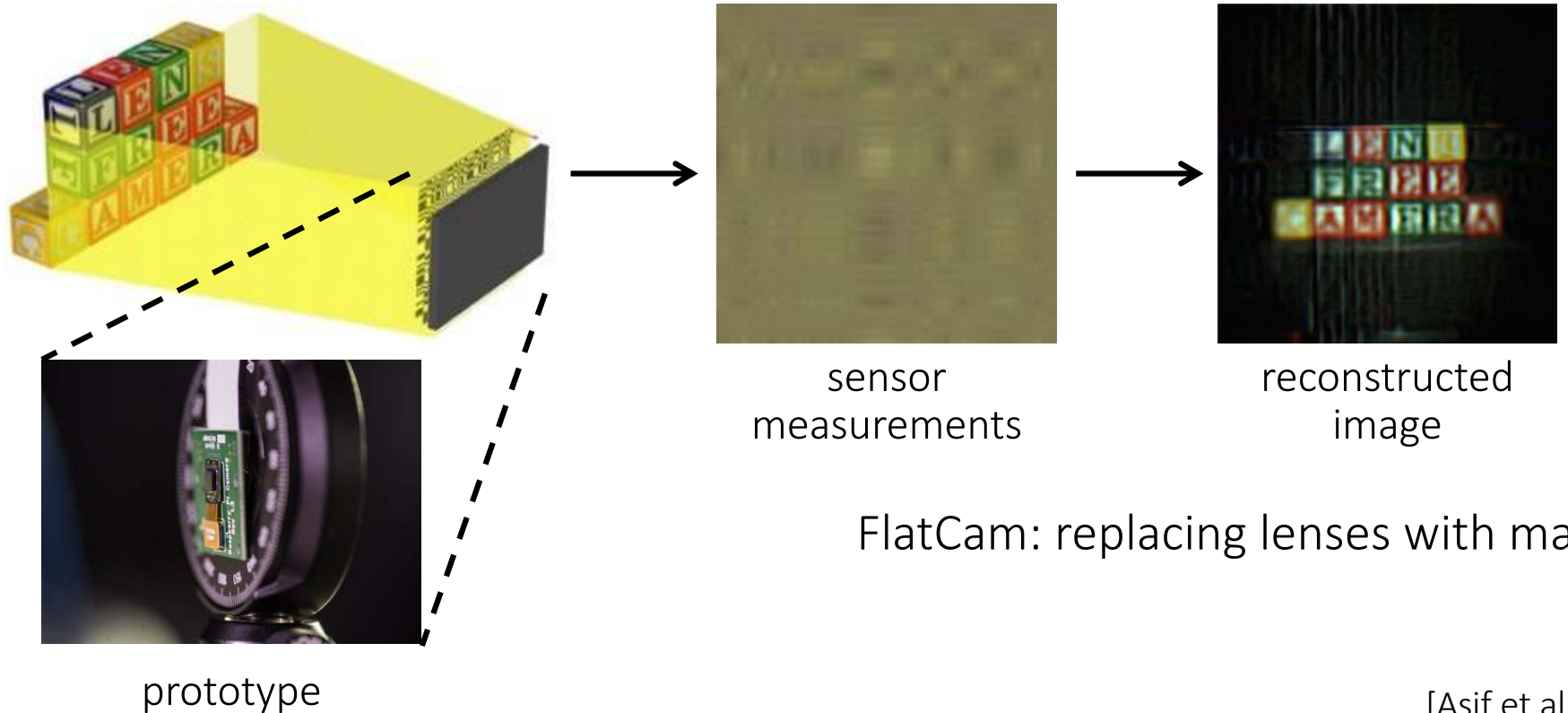


inferred depth

Coded aperture for single-image depth and refocusing

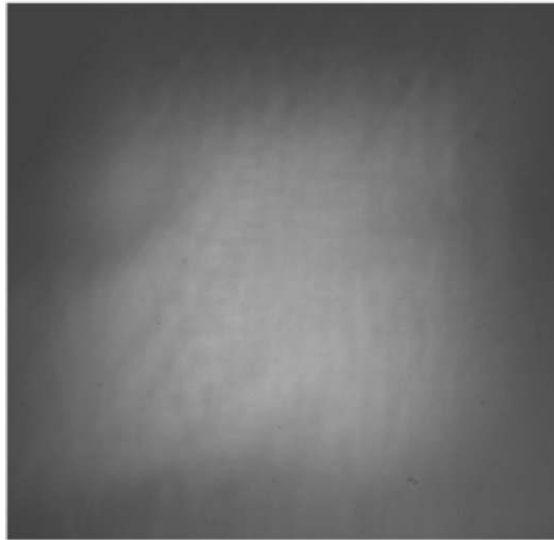
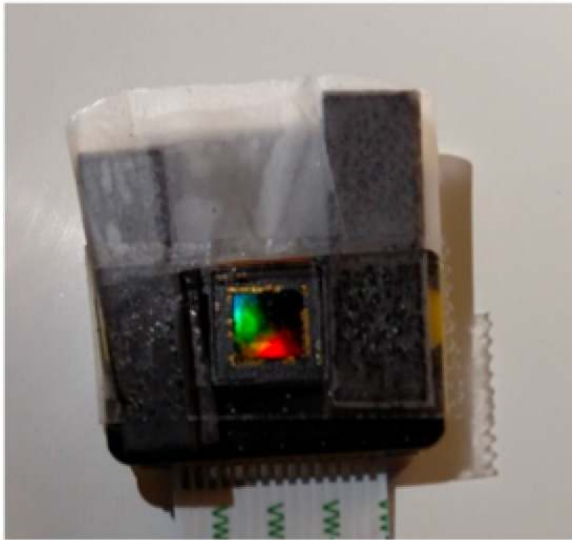
[Levin et al., SIGGRAPH 2007]

Remove Lenses all Together!



[Asif et al. 2015]

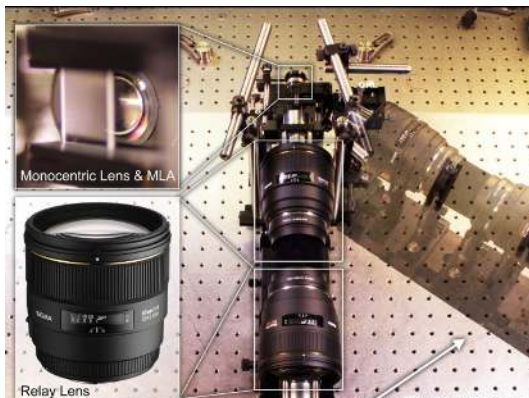
Scotch Tape DiffuserCam!



One of your
homeworks!

[Kuo, Antipa, Biscarrat, Parthasarathy]

Computational Cameras



generalized optics
between scene and sensor

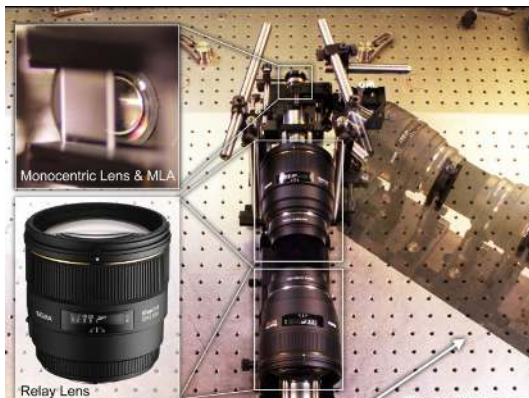


digital sensor to capture focused
light (electrical process)

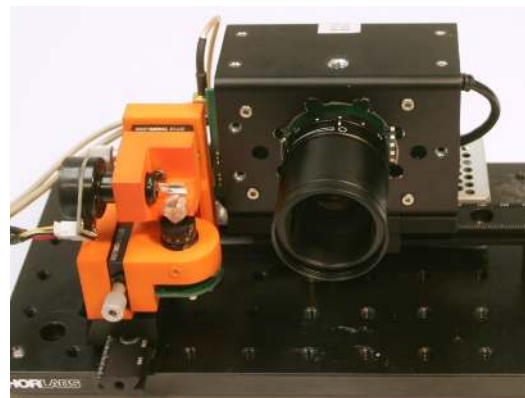


arbitrary computation
between sensor and image

Computational Cameras



generalized optics
between scene and sensor



unconventional sensing
and illumination



arbitrary computation
between sensor and image

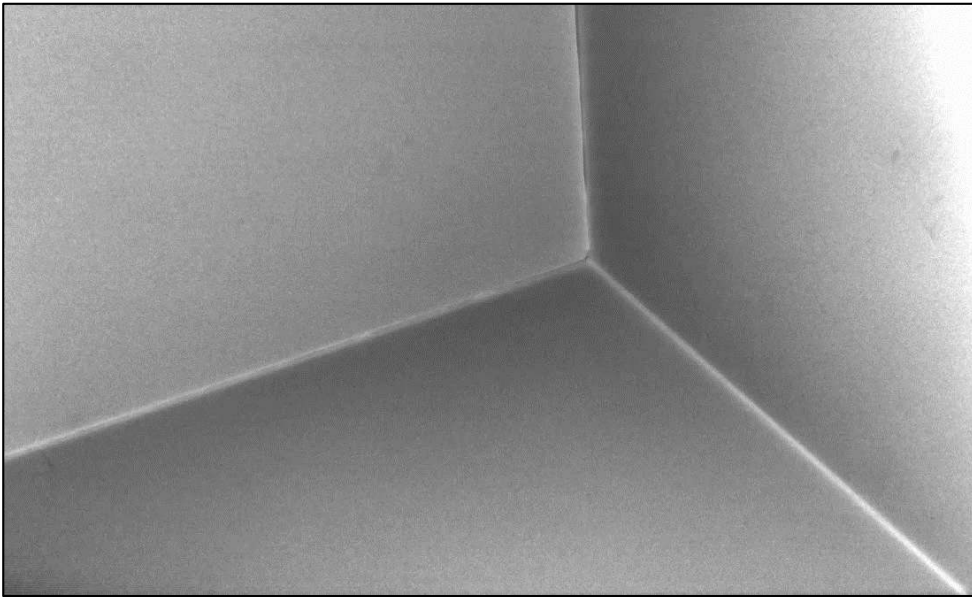
Slide courtesy of Ioannis Gkioulekas

Structured Light for Epipolar Imaging

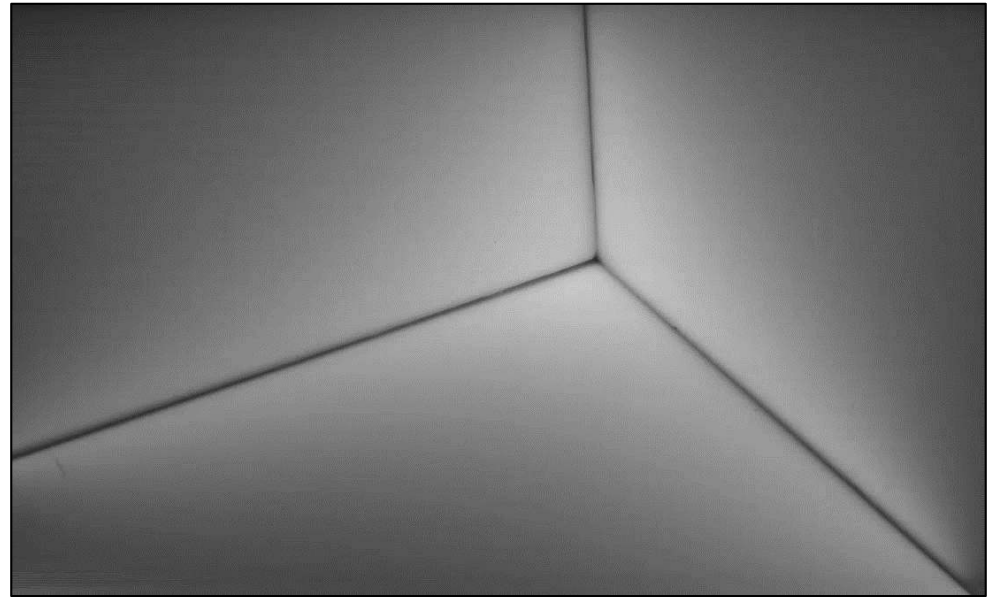


[O'Toole et al., SIGGRAPH 2015]

Structured Light for Epipolar Imaging



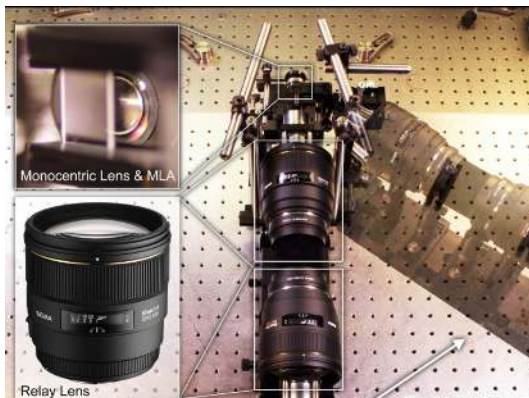
direct photons



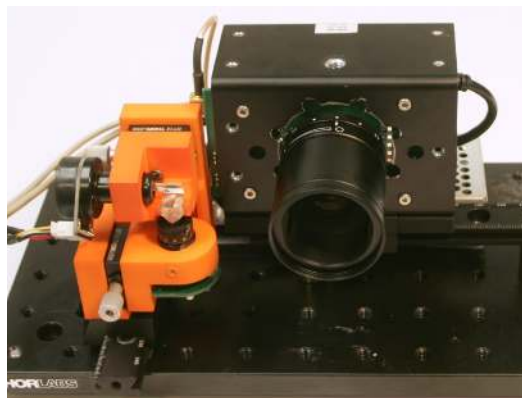
indirect photons

[O'Toole et al., SIGGRAPH 2015]

Computational Cameras



generalized optics
between scene and sensor

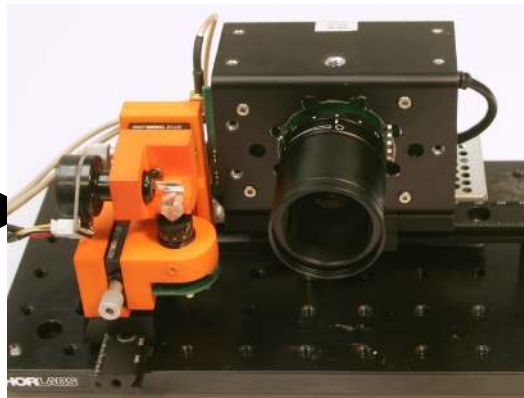
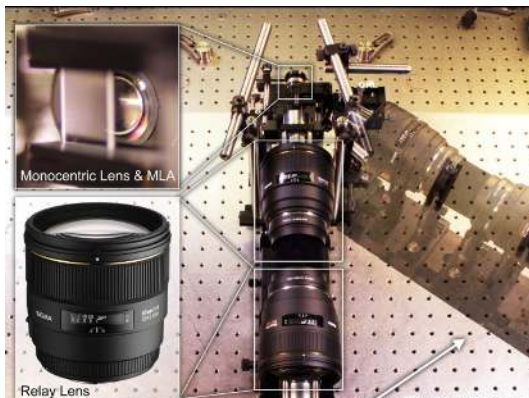


unconventional sensing
and illumination



arbitrary computation
between sensor and image

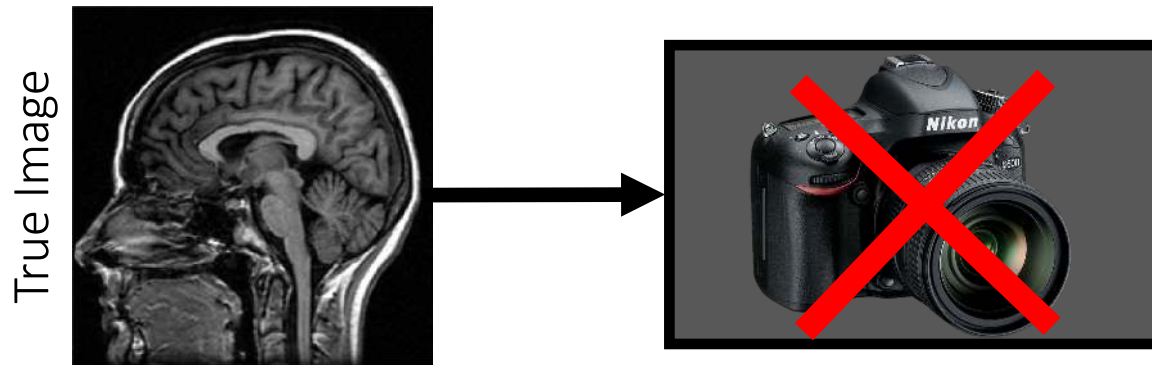
Computational Cameras



Reconstruction



Computational Cameras

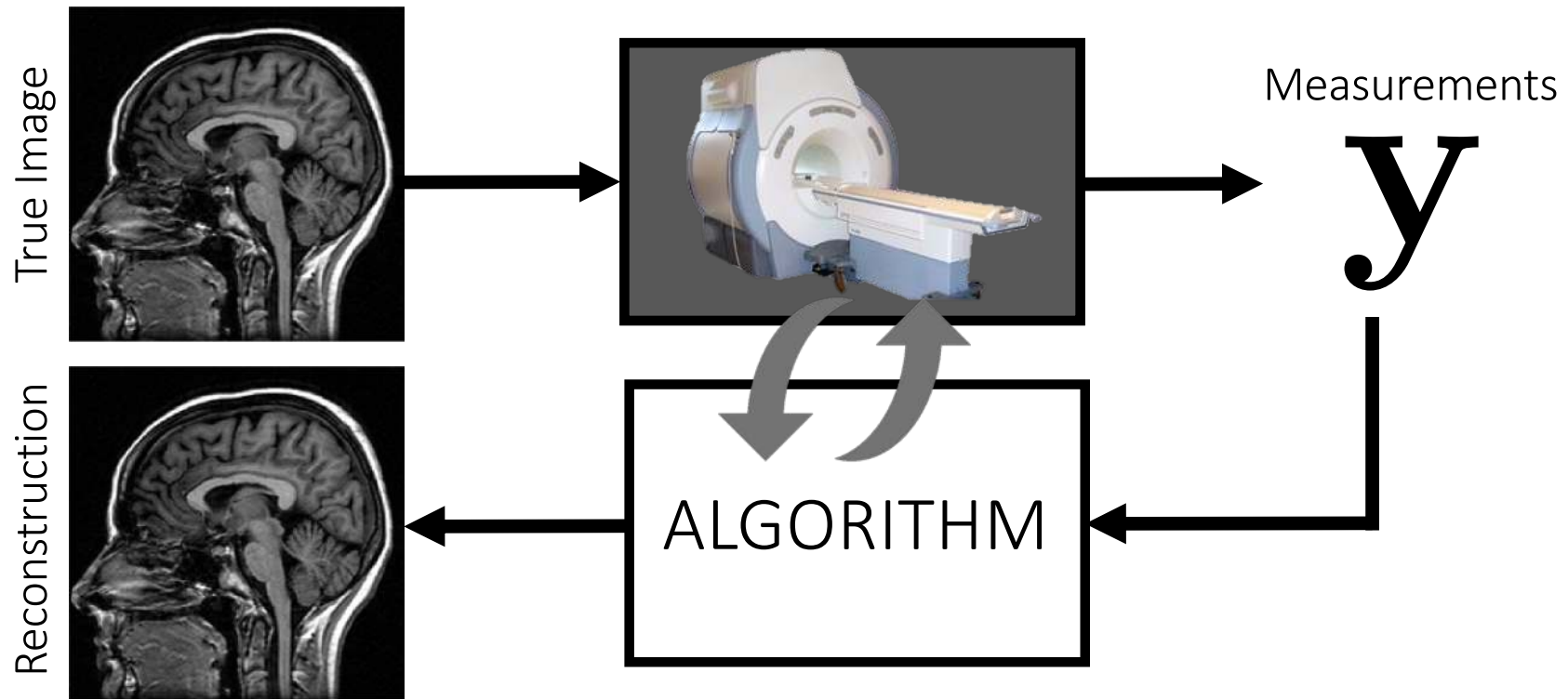


Cannot Use a Standard Camera to See “Invisible” Scenes

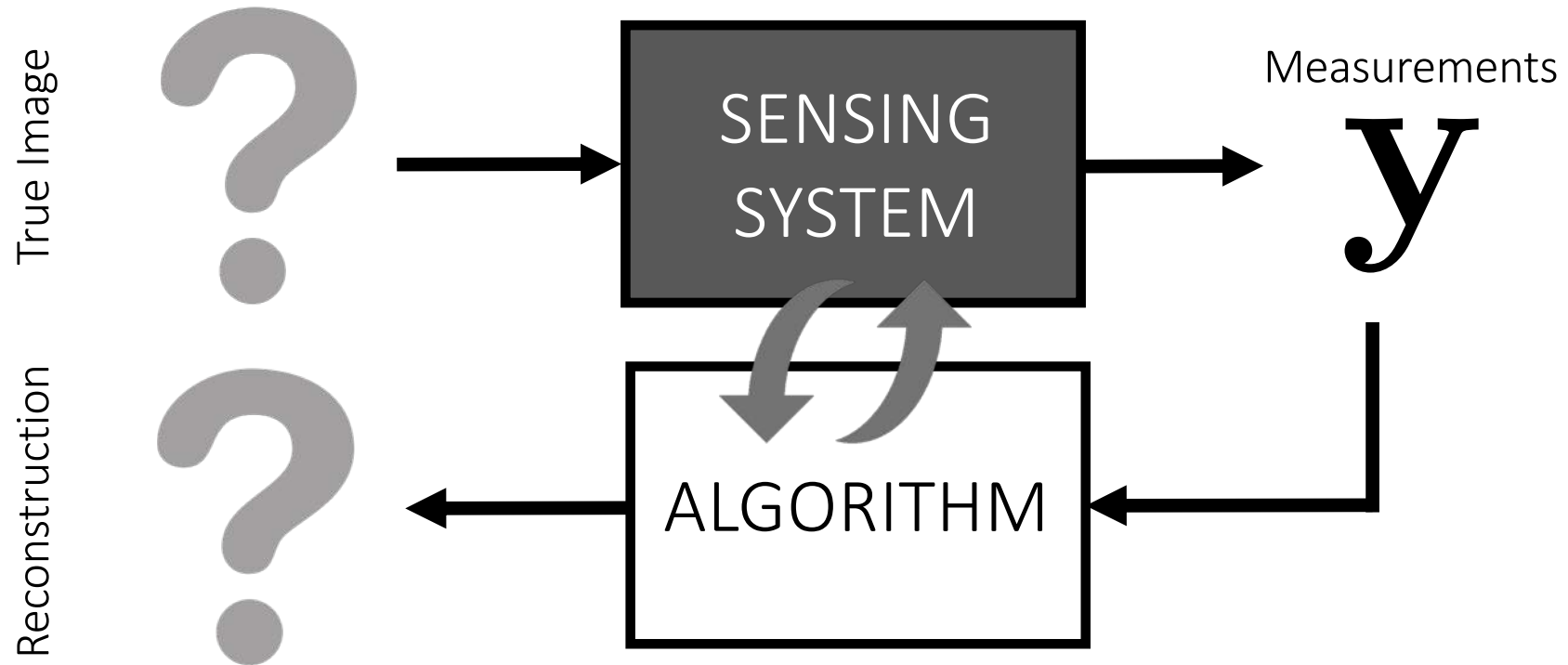
Computational Cameras



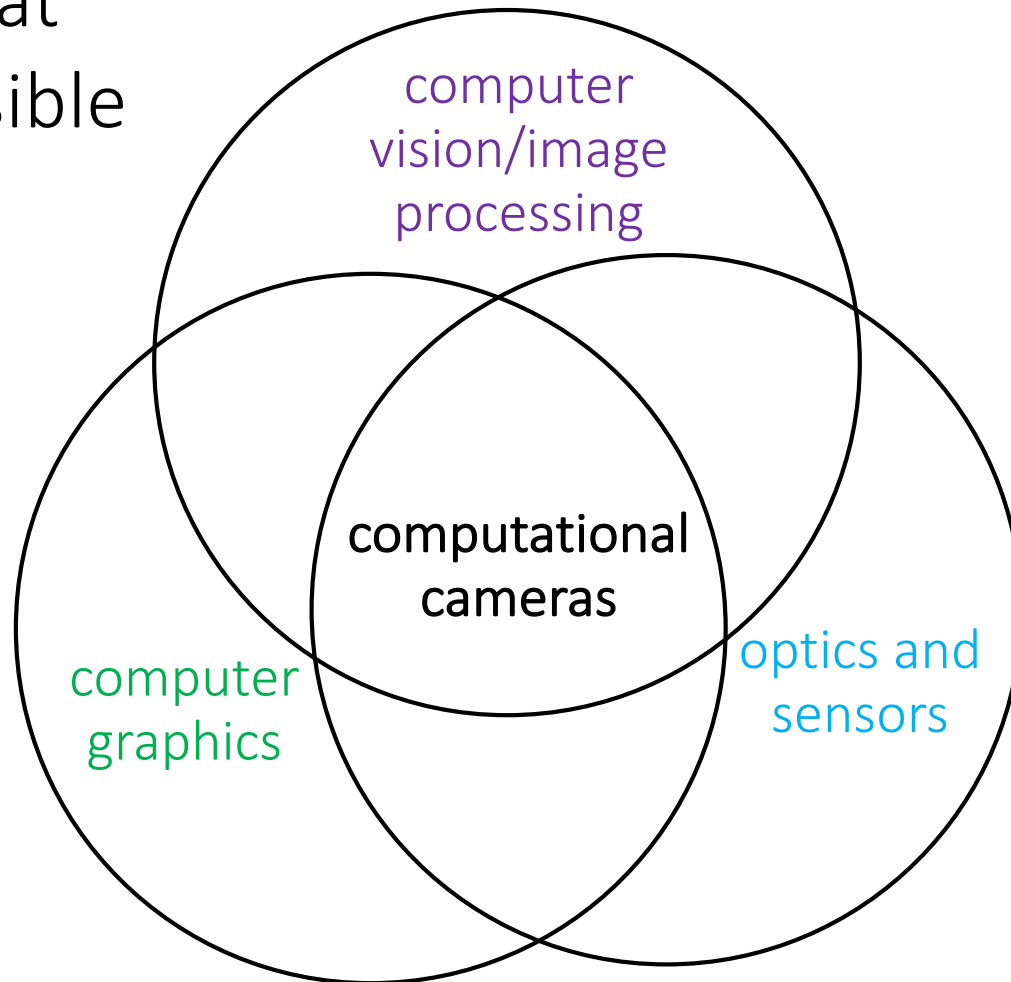
Computational Cameras



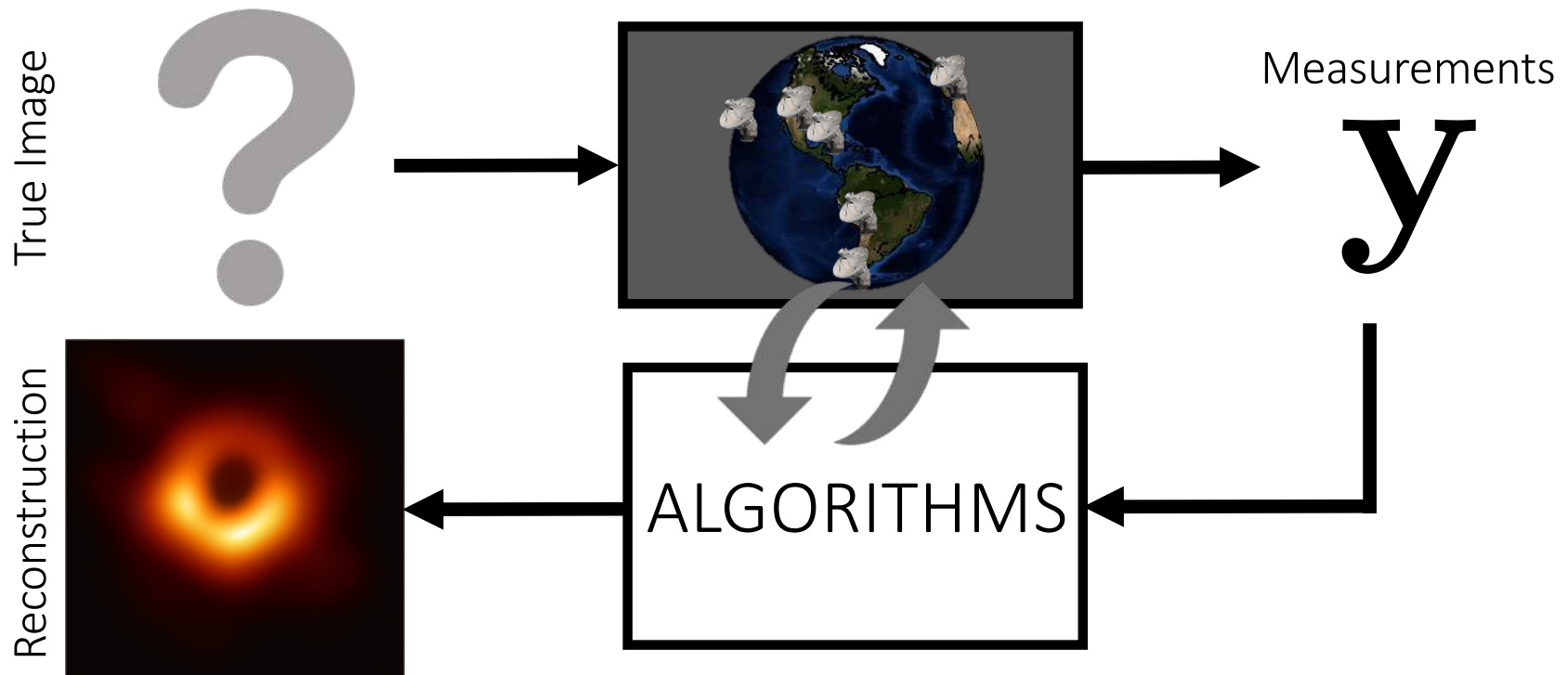
Computational Cameras



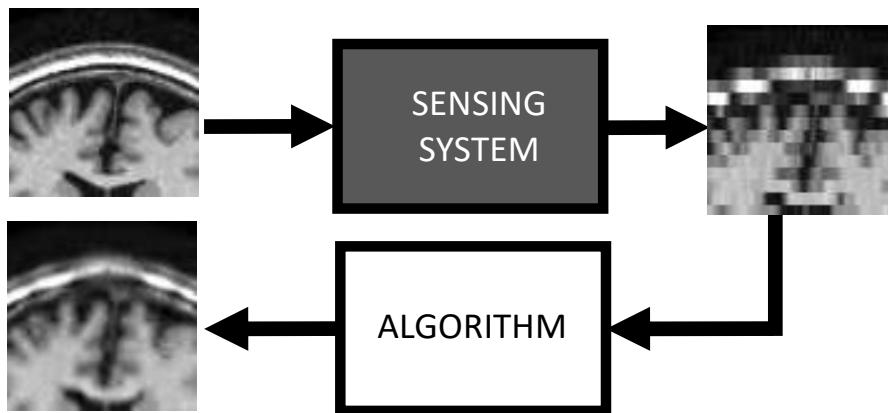
Systems that See the Invisible



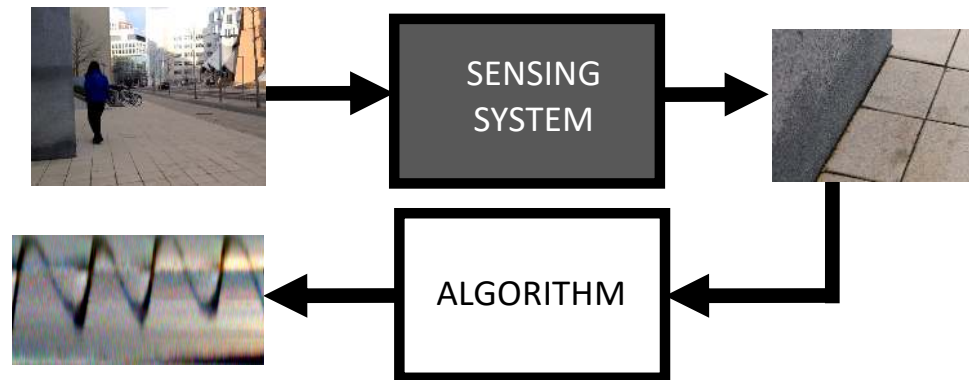
Unconventional Imaging to Recover the Invisible



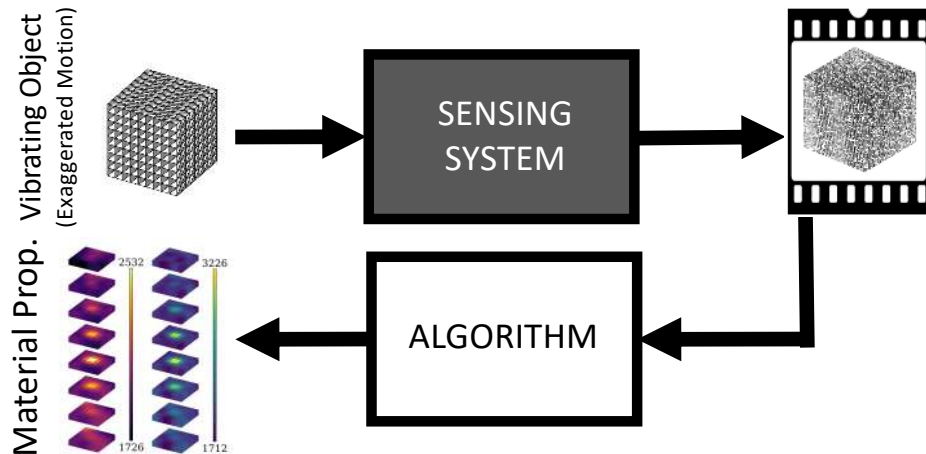
Brain MRI Super-Resolution



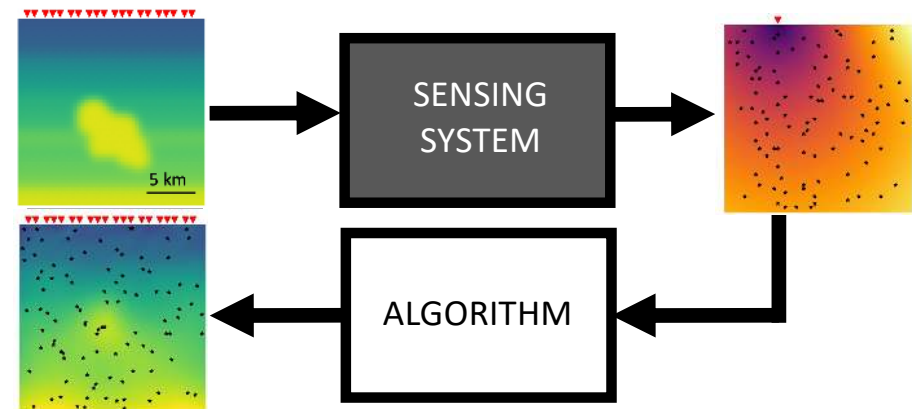
Seeing Behind Corners



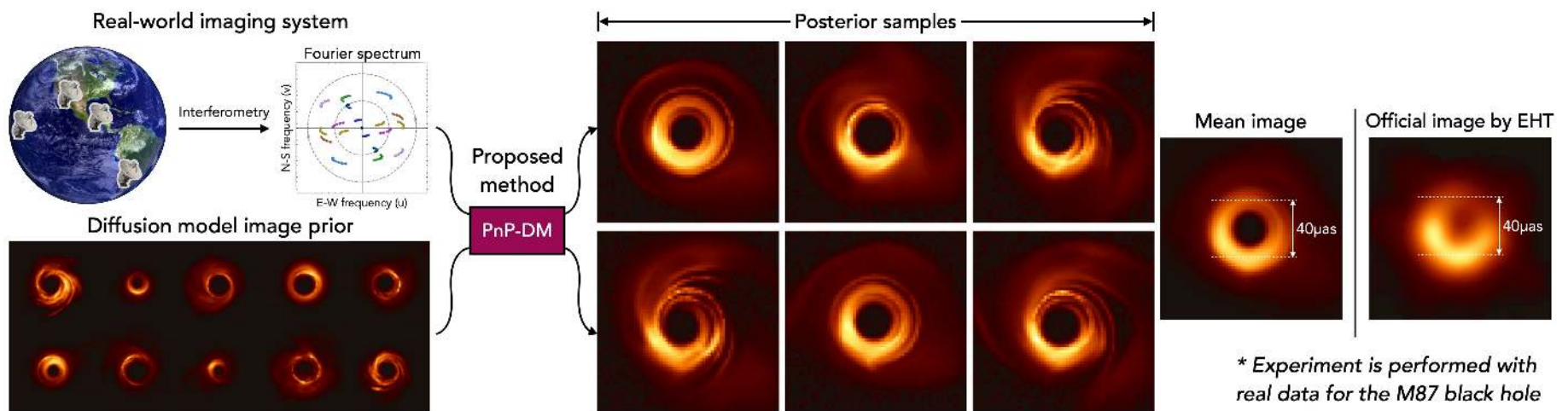
Recovering Material Properties



Seismic Tomography



Using Expected Structure to Solve for the Image



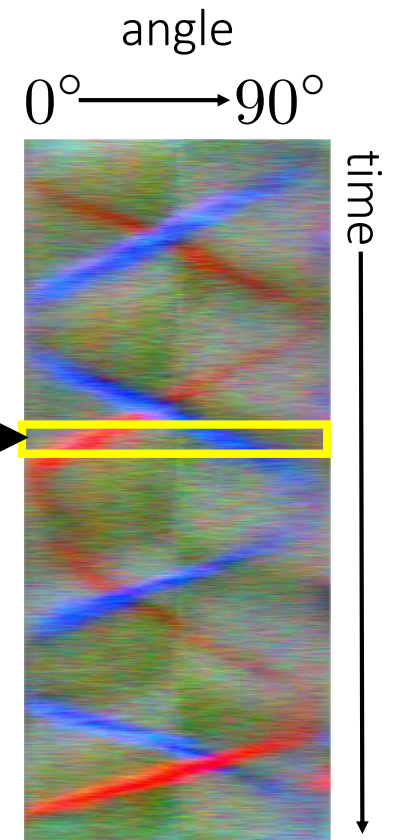
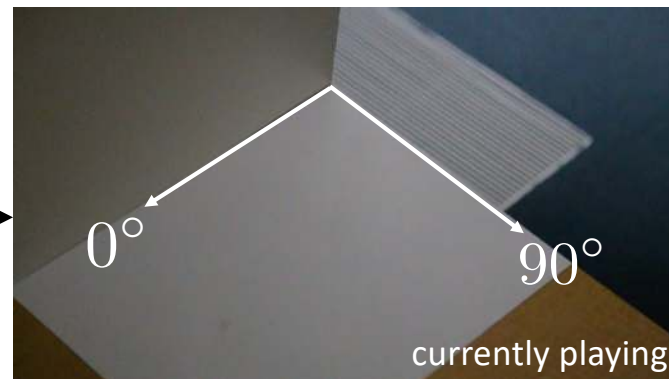
[Wu, et al, 2024]

Seeing behind corners

Hidden Scene



Video of Corner



One of your
homeworks!

[Bouman et al, 2017]

Course Schedule

<i>Week</i>	<i>Date</i>	<i>Lecture Topic</i>	<i>Homework</i>
<i>1</i>	<i>March 31</i>	<i>Syllabus & Computational Cameras</i>	
<i>1</i>	<i>April 2</i>	<i>Color</i>	
<i>2</i>	<i>April 7</i>	<i>Pinholes & Lenses</i>	
<i>2</i>	<i>April 9</i>	<i>Digital Photography</i>	<i>HW 1 released</i>
<i>3</i>	<i>April 14</i>	<i>Noise & Filtering</i>	
<i>3</i>	<i>April 16</i>	<i>Exposure & High Dynamic Range Photography</i>	<i>HW 1 due HW 2 released</i>
<i>4</i>	<i>April 21</i>	<i>Pyramids & Blending</i>	
<i>4</i>	<i>April 23</i>	<i>NO CLASS</i>	

Course Schedule

5	April 28	Light Fields & Focal Stacks	HW 2 due HW 3 released
5	April 30	Signal Processing & Linear Spatial Filters	
6	May 5	Linear Inverse Problems & Deconvolution	HW 3 due HW 4 released
6	May 7	Coded Aperture & Lensless Cameras	
7	May 12	Compressed Sensing & Single Pixel Cameras	HW 4 & final project proposal and literature review due HW 5 released
7	May 14	Radio Interferometry & Black Hole Imaging	

Course Schedule

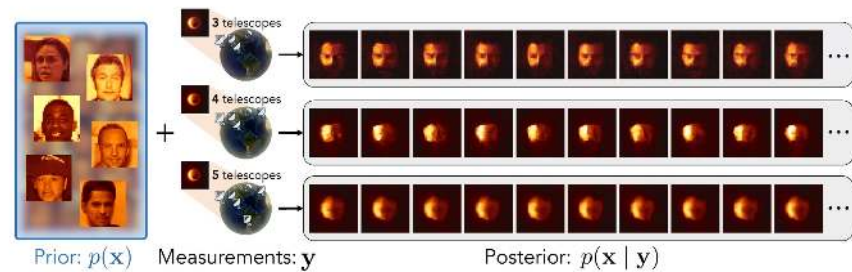
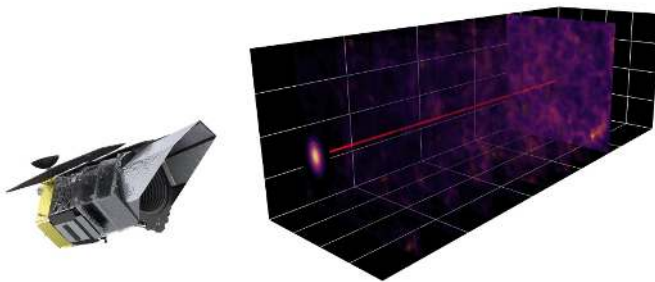
8	May 19	<i>Deep Learning for Inverse Problems (guest lecture by Berthy Feng)</i>	<i>HW 5 due</i>
8	May 21	<i>Tomography & Single-view 3D Reconstruction (guest lecture by Brandon Zhao)</i>	
9	May 26	<i>Phase Retrieval & Applications (guest lecture by Jason Wang and Rodrigo Ferrer-Chavez)</i>	
9	May 28	<i>Diffraction Imaging (guest lecture by Hosea Nelson and Dmitry Eremin)</i>	<i>Interim project report due</i>
10	June 2	<i>Finals Week – No Lecture</i>	
10	June 4	<i>Finals Week – No Lecture</i>	<i>Final projects and presentations due</i>

Invited Speakers

Brandon Zhao (Caltech)



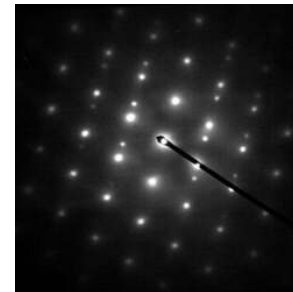
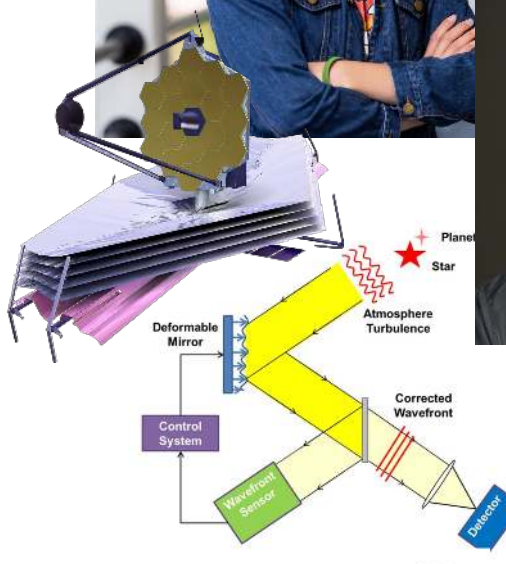
Dr. Berthy Feng (Caltech->MIT)



Invited Speakers

Prof. Jason Wang and Rodrigo Ferrer-Chavez (Northwestern)

Prof. Hosea Nelson and Dr. Dmitry Eremin (Caltech)



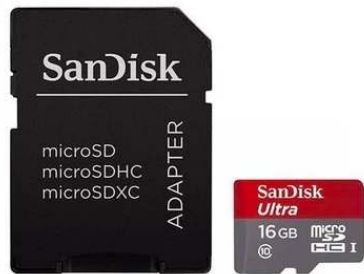
Questions?

Homework

- HW1: Color, Optics, & Digital Camera Pipeline
- HW2: Filtering, High Dynamic Range Imaging & Tonemapping
- HW3: Blending & Light Field Imaging
- HW4: Inverse Problems & the Corner Camera
- HW5: Lensless “Scotch Tape” Camera (break your camera!)
- Final Project

1.5 weeks for this one!

Supplies



Supplies – HW1 (due April 16)



Pinhole Camera



Questions?